

TYABO SALIHU

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Professional Summary

Senior Data Engineer with 7+ years of experience building large-scale data platforms across financial services, healthcare, and e-commerce. Deep hands-on work designing and operating cloud-native pipelines and warehouses separately on AWS, Azure, and GCP, enabling analytics, BI, and ML teams with reliable, secure, and well-modeled data. Strong record partnering with risk, clinical, and product stakeholders to translate complex multi-domain requirements into scalable architectures and measurable impact.

Core Skills

Languages: Python, SQL, Bash

Data Processing: Apache Spark, PySpark, Kafka/Kinesis/Pub/Sub, Hadoop

Cloud Platforms: AWS: S3, EMR, Glue, Redshift, Lambda, IAM

Azure: Data Lake Storage, Synapse Analytics, Azure Data Factory, Azure Databricks

GCP: BigQuery, Cloud Storage, Dataflow/Dataproc, Pub/Sub

Warehousing: Snowflake, Redshift, BigQuery, Synapse, Delta Lake/Lakehouse

Orchestration & ELT: Airflow, dbt, Azure Data Factory, Cloud Composer

Databases: PostgreSQL, MySQL/SQL Server, NoSQL (DynamoDB, Cosmos DB, Bigtable)

Practices: ETL/ELT, data modeling (star/snowflake), data quality, lineage, governance, security, cost optimization

Professional Experience

Wayfair

Senior Data Engineer – E-commerce

May 2023 – Present

- Design, build, and maintain batch and streaming data pipelines on AWS using S3, EMR, Glue, and Lambda to process clickstream, product, cart, and order data.
- Implement dimensional and wide-table models in Redshift/Snowflake on AWS to support product analytics, merchandising, and marketing attribution.
- Develop ELT workflows orchestrated by Airflow (or native orchestrators) with clear dependencies, retries, SLAs, and automated data-quality checks.
- Optimize Spark clusters on EMR and complex SQL queries on Redshift (distribution keys, sort keys, compression, workload management) to handle multi-TB datasets.
- Build and maintain centralized event schemas and tracking standards for web and app telemetry, ensuring consistent analytics across products.
- Partner with data scientists to deliver feature pipelines for recommendation systems, search ranking, and personalization services, including versioned feature sets.
- Implement monitoring and alerting for pipeline health using CloudWatch and custom metrics, reducing time to detect and resolve failures.
- Collaborate with security and compliance to enforce IAM policies, encryption at rest and in transit, and tokenization/masking for sensitive customer and payment data.
- Introduce standardized logging, error-handling, and replay mechanisms for critical pipelines, simplifying troubleshooting and backfills.
- Lead design and code reviews, improve coding standards, and mentor junior engineers on AWS-based data architectures and best practices.

Texas Trust Credit Union

Data Engineer *Mar 2021 – May 2023*

- Modeled accounts, transactions, delinquency, collections, and risk exposure in Azure Synapse Analytics, supporting regulatory, finance, and risk reporting.
- Built and maintained data pipelines on Azure Data Factory and Azure Databricks to ingest data from core banking systems, payment processors, credit bureaus, and CRM into Azure Data Lake Storage.
- Implemented incremental and CDC-based pipelines (using change tables, log capture, and watermarking) to keep risk and compliance data up to date with minimal batch windows.
- Tuned Spark jobs in Databricks and SQL queries in Synapse (partitioning, indexing, caching, materialized views) to meet nightly and month-end reporting deadlines.
- Developed reusable data-quality frameworks (reconciliations, threshold checks, exception tables) and integrated them into Data Factory pipelines.
- Worked closely with risk analytics and modeling teams to deliver curated, documented datasets for portfolio analysis, stress testing, and scorecard development.
- Collaborated with InfoSec to implement role-based access control using Azure AD groups and managed identities, including segregation of duties.
- Contributed to infrastructure-as-code for data components (e.g., Bicep/ARM/Terraform for Synapse, Data Factory, Databricks), enabling consistent, repeatable deployments.
- Defined and documented data lineage from upstream transactional systems to downstream risk dashboards and regulatory reports to support audits.
- Participated in on-call rotations for critical Azure pipelines, performing root-cause analysis, implementing long-term fixes, and documenting incident runbooks.

Optum

Data Engineer

Feb 2019 – Feb 2021

- Developed ETL/ELT pipelines on GCP using BigQuery, Dataflow, and Cloud Storage to integrate claims, encounters, EMR, provider, and member data.
- Designed normalized and dimensional schemas in BigQuery for patients, providers, facilities, diagnoses, procedures, and episodes of care to support outcomes and utilization analytics.
- Implemented HIPAA-compliant data handling, including de-identification pipelines, access-controlled views, and encryption using CMEK keys in GCP.
- Built reusable templates in Dataflow/Dataproc for ingesting and transforming large healthcare datasets, with built-in validation and logging.
- Partnered with clinical analytics and operations teams to convert metric definitions (readmissions, utilization rates, quality scores) into robust SQL transformations.
- Created healthcare-specific data marts (e.g., readmission analysis, chronic condition cohorts, risk-stratification) for BI teams and downstream applications.
- Established data-quality checks such as coding validation (ICD/CPT), referential integrity, time-window checks, and anomaly detection for claims and clinical data.
- Worked with platform teams to optimize BigQuery costs (partitioned/clustered tables, reservation management, materialized views) while maintaining performance.
- Documented data models, table contracts, SLAs, and lineage in internal catalogs to improve discoverability and trust for analysts and data scientists.
- Supported migration from on-prem healthcare warehouses to GCP, including planning, mapping, and re-platforming ETL pipelines.

Education

Bsc Computer Science 2013

Anglia Ruskin University Chelmsford UK